



RevoFin – Loan Performance Analysis

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Project Background

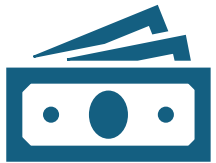


RevoFin is a fintech company focused on transforming the lending experience through technology and data-driven insights. It provides an online platform that connects individuals and small businesses with various loan options, using automated credit assessments and streamlined application processes to deliver personalized and competitive financial solutions.



The company generates revenue through transaction fees, loan interest, and strategic partnerships. Within this setup, the data analyst plays a key role in driving growth by analyzing customer behavior, loan performance, and market trends to optimize lending strategies and improve overall business performance.

Project Objective



Present the total outstanding loan amounts across the portfolio.



Identify and highlight key risk metrics related to the loan portfolio.



Deliver a detailed analysis of outstanding loans and associated risks based on:

Loan characteristics (e.g., loan type, tenure, interest rate)

Customer attributes (e.g., income level, employment status, home ownership)

Overall, RevoFin maintains strong portfolio health with high repayment rates (98.73% TKB30)

270,299

Total Customer

US\$ 4,166,072,400

OutStanding (OS)

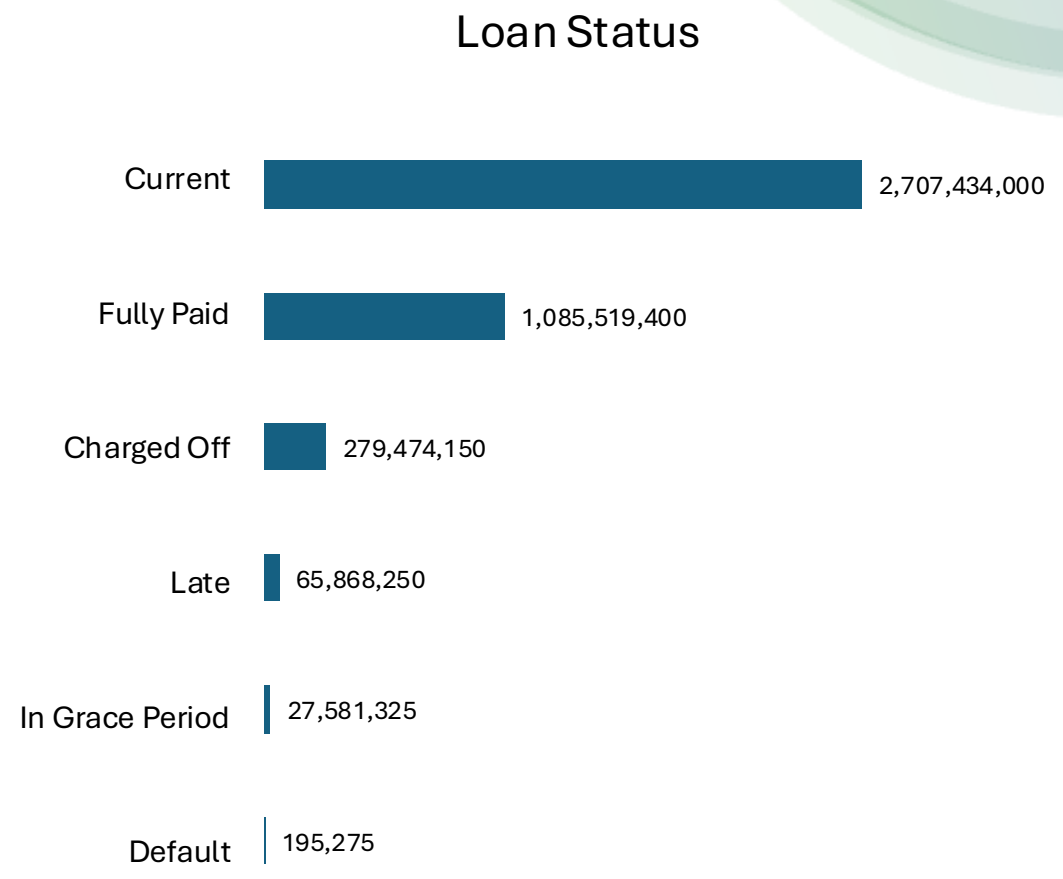
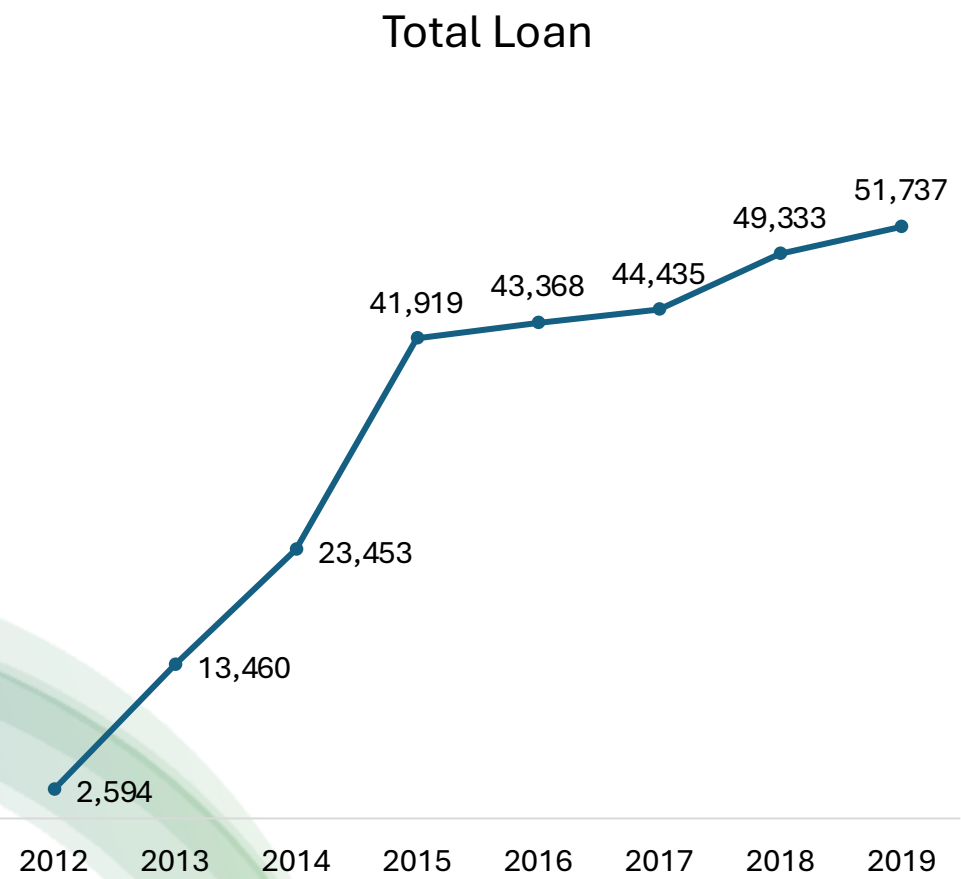
US\$ 4,165,877,125

ENR

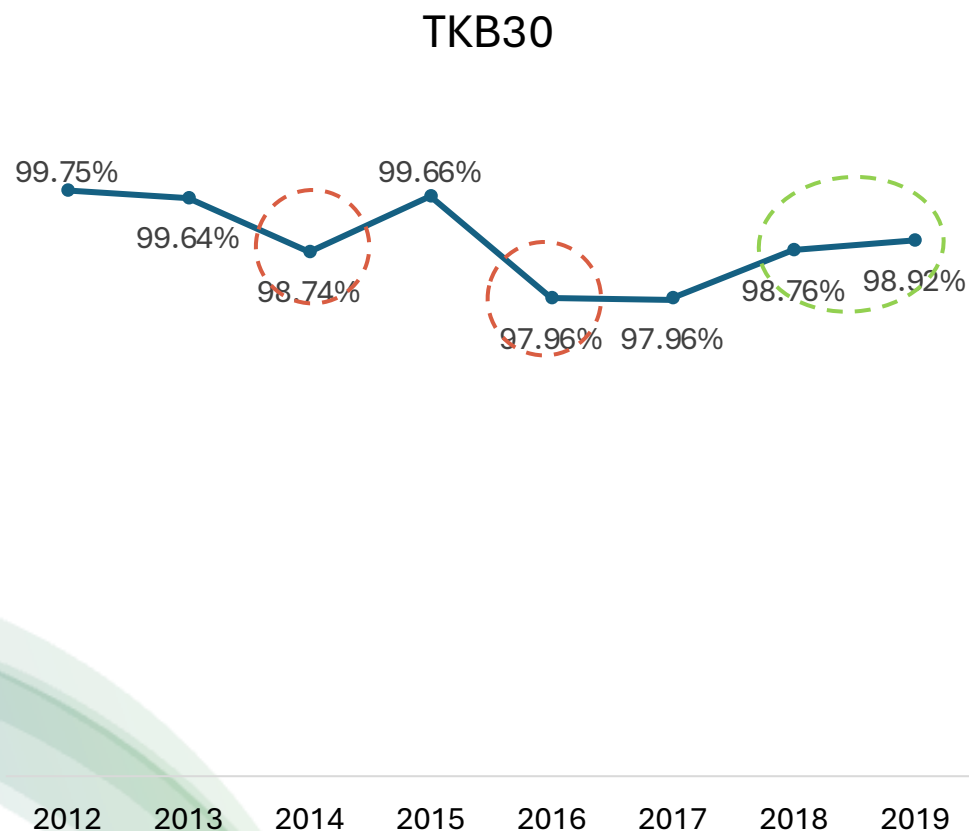
98.73%

TKB30

Consistent growth in loan volume with majority of loans are fully repaid



Portfolio performance deteriorated in 2014 and 2016 before recovering in later cohorts



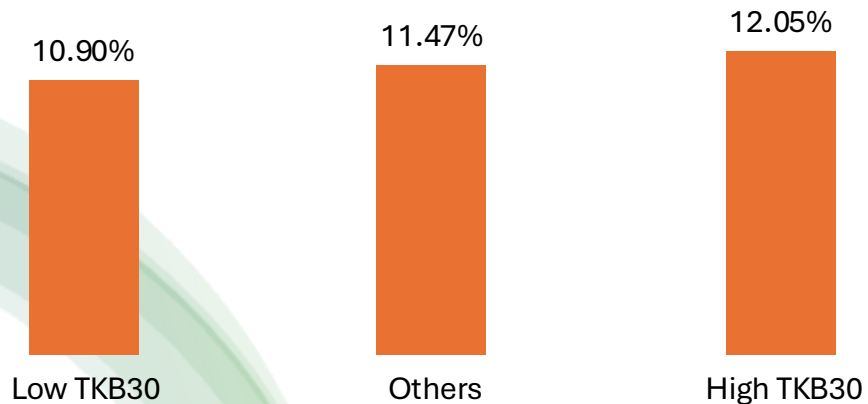
For deeper analysis, we will dive using Cohort Analysis with grouping:

- Low TKB30: 2014 and 2016
- High TKB30: 2018 and 2019
- Others: 2015 and 2017

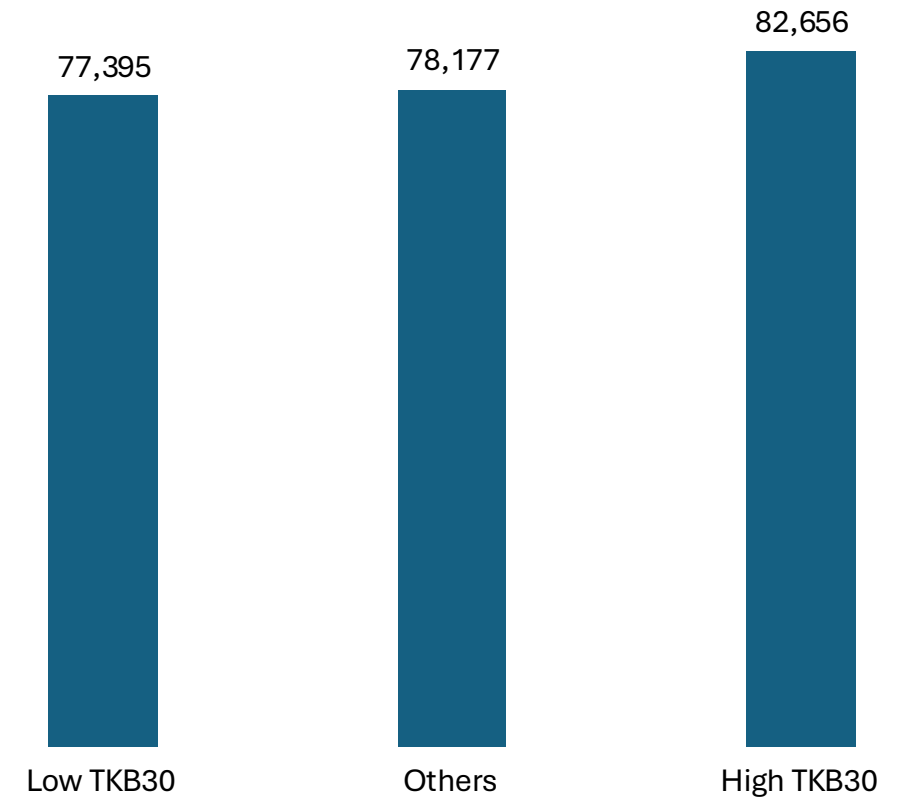
(* Excluding 2012-2013 from the Cohort Group Analysis because total loan takers are low.)

Lower income and lack of home ownership are correlated factors of poor repayment performance

Home Ownership

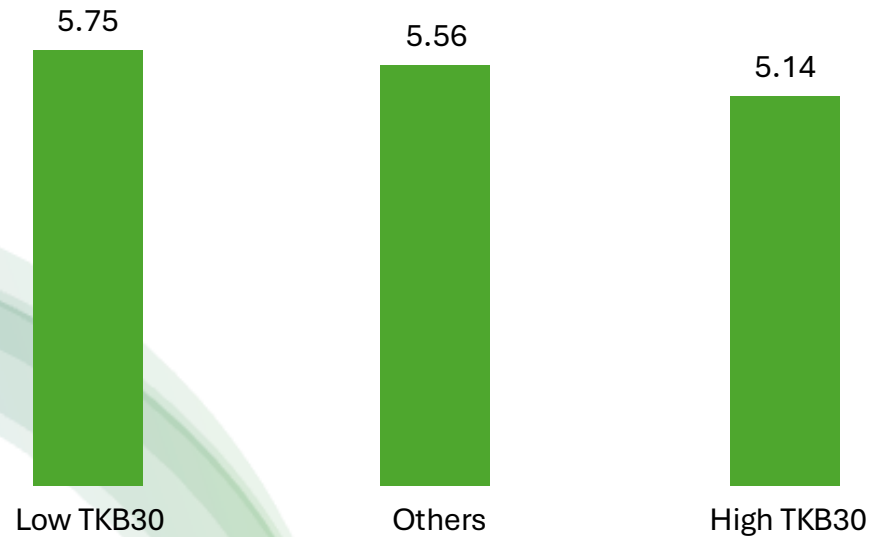


Average Income

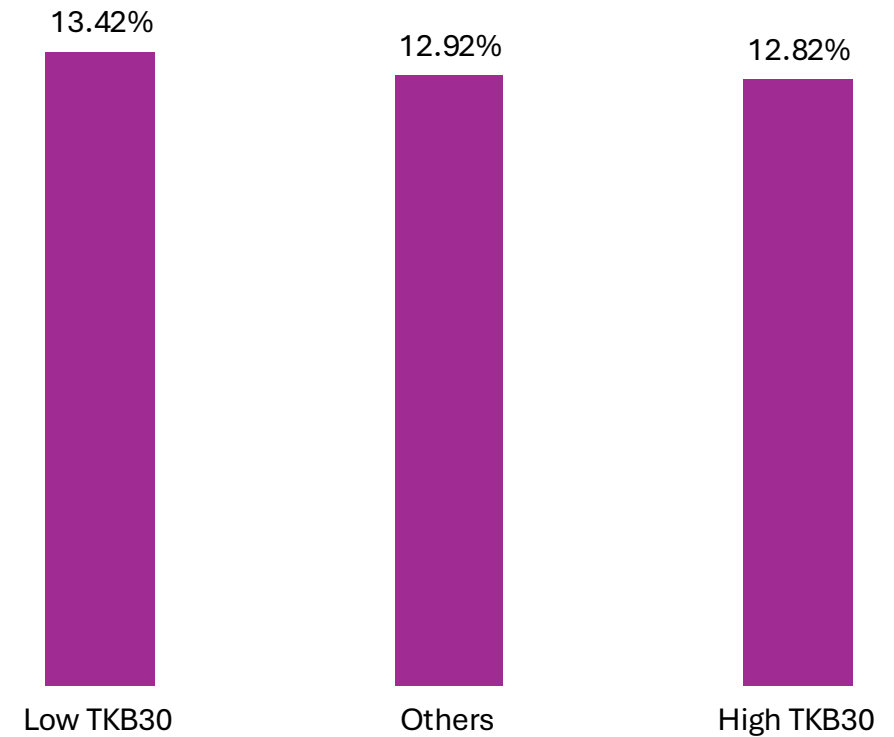


Lower installment burden tend to correlate with better repayment performance

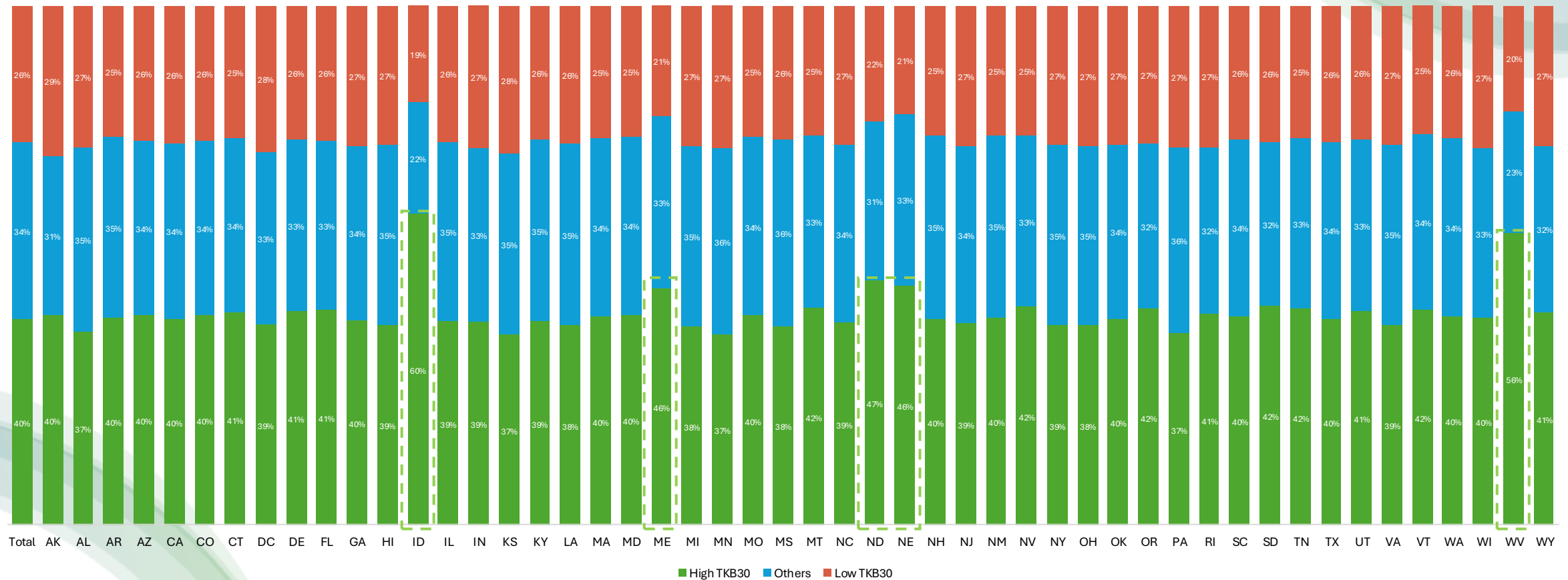
Employment Length



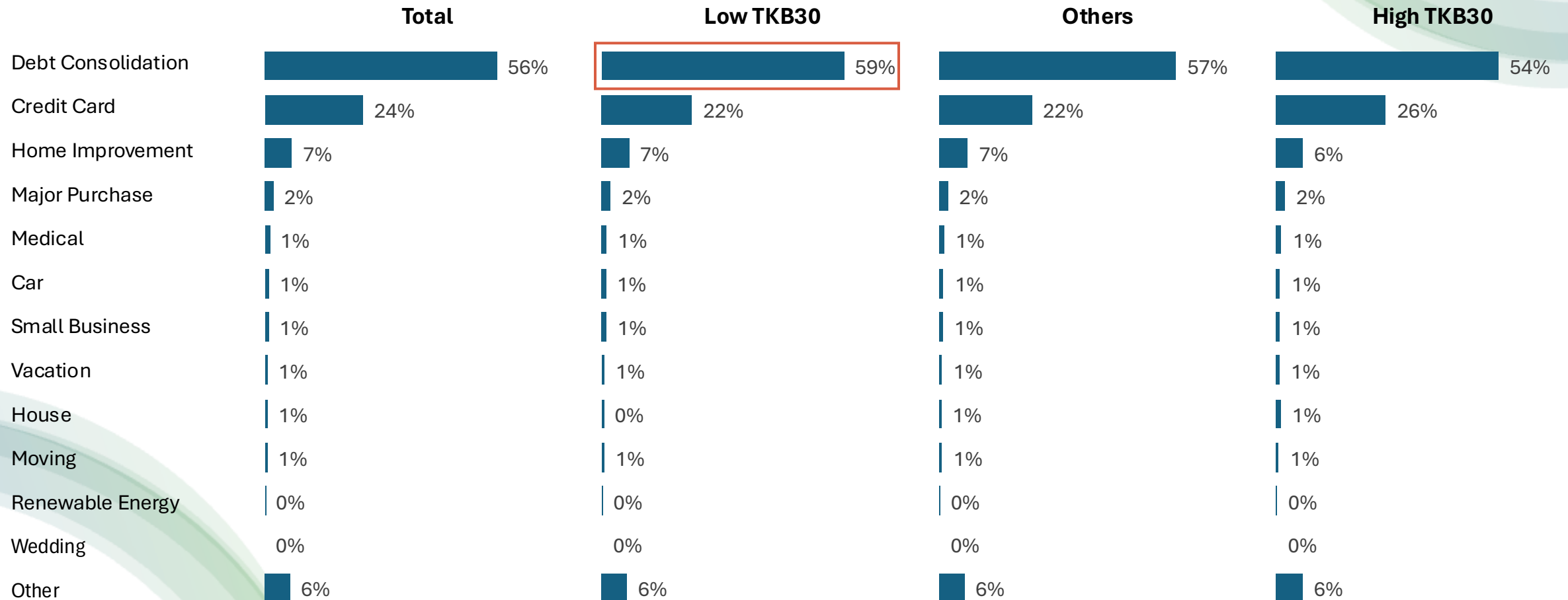
Installment Rate



High-Performing Loans are concentrated in specific states such as Idaho and West Virginia followed by North Dakota, Maine, and Nebraska



Debt consolidation loans exhibit higher delinquency risk



Recommendation

Observation	Business Impact	Isolation	Prioritization	Recommendation
Low-performing loans are concentrated among borrowers with income below \$80K, no home ownership, longer employment tenure, and debt consolidation purposes	Higher delinquency directly reduces RevoFin's net interest income and increases provisioning costs	Loan application criteria	High – directly prevents TKB30 from declining further	Apply a multi-factor risk flag at application stage: automatically trigger manual review for applicants meeting 3 or more of these criteria — income <\$80K, non-homeowner, employment >5 years, and debt consolidation purpose
Higher installment-to-income ratio correlates with lower repayment performance across all cohorts	Reducing installment rates lowers revenue per loan, but reduces charge-off losses	Instalment rate policy	Medium – improves repayment for existing at-risk borrowers	Introduce an on-time payment reward: offer a 0.5–1% installment rate reduction after 6 consecutive on-time payments, targeting borrowers in the high-risk segment to incentivize continued repayment
High-performing loans are disproportionately concentrated in Idaho, West Virginia, North Dakota, Maine, and Nebraska	Underperforming states drag down the national TKB30 and signal geographic risk concentration	State-level analysis	Low – a longer-term portfolio optimization regulation	Monitor the bottom 10 states by TKB30 quarterly; consider tightening credit criteria or reducing loan limits in persistently underperforming states before expanding volume there

Appendix

SQL – Q1

Total Customer

```
SELECT  
array_to_string(regexp_extract_all(customer  
id, "[a-zA-Z0-9]", "")) as customer_id  
FROM `nextproject-001.FIP.customer`
```

SQL – Q2

Loan Status

```
SELECT loan_status, COUNT(customer_id) AS  
total_count  
FROM `nextproject-001.FIP.loan`  
GROUP BY 1  
ORDER BY 2 DESC
```

SQL – Q3

Total Customer

```
SELECT COUNT(DISTINCT customer_id)
FROM `nextproject-001.FIP.loan`
```

ENR

```
SELECT SUM(loan_amount)
FROM `nextproject-001.FIP.loan`
WHERE loan_status != 'Default'
```

OS

```
SELECT SUM(loan_amount)
FROM `nextproject-001.FIP.loan`
```

SQL – Q4

TKB30

```
WITH cte1 AS (SELECT loan_id, loan_amount  
FROM `nextproject-001.FIP.loan`  
WHERE loan_status = 'Late (31-120 days)'),
```

```
cte2 AS (SELECT loan_id, loan_amount  
FROM `nextproject-001.FIP.loan`)
```

```
SELECT 1 - (SUM(cte1.loan_amount) /  
SUM(cte2.loan_amount))  
FROM cte1  
RIGHT JOIN cte2  
ON cte1.loan_id = cte2.loan_id
```

SQL – Q5

TKB30 per cohort

```
WITH cte1 AS (SELECT loan_id, loan_amount
FROM `nextproject-001.FIP.loan`
WHERE loan_status = 'Late (31-120 days)'),
```

```
cte2 AS (SELECT loan_id, issue_year,
loan_amount
FROM `nextproject-001.FIP.loan`)
```

```
SELECT cte2.issue_year, 1 -
(SUM(cte1.loan_amount) /
SUM(cte2.loan_amount))
FROM cte1
RIGHT JOIN cte2
ON cte1.loan_id = cte2.loan_id
GROUP BY 1
ORDER BY 1
```

SQL – Q7 – a, b, c, f

```
WITH cte1 AS (  
  SELECT  
    issue_year,  
    c.customer_id,  
    annual_inc,  
    IF(home_ownership = 'OWN', 1, 0) AS home_ownership,  
    CASE  
      WHEN emp_length = '1 year' THEN 1  
      WHEN emp_length = '2 years' THEN 2  
      WHEN emp_length = '3 years' THEN 3  
      WHEN emp_length = '4 years' THEN 4  
      WHEN emp_length = '5 years' THEN 5  
      WHEN emp_length = '6 years' THEN 6  
      WHEN emp_length = '7 years' THEN 7  
      WHEN emp_length = '8 years' THEN 8  
      WHEN emp_length = '9 years' THEN 9  
      WHEN emp_length = '10+ years' THEN 10  
      ELSE 0 END AS num_enp_length,  
    int_rate  
  FROM `nextproject-001.FIP.loan` l  
  LEFT JOIN `nextproject-001.FIP.customer` c  
  ON l.customer_id = c.customer_id),
```

```
cte2 AS (  
  SELECT  
    issue_year,  
    COUNT(customer_id) AS total_customer,  
    AVG(annual_inc) AS avg_income,  
    SUM(home_ownership) AS home_ownership,  
    AVG(num_enp_length) AS avg_enp_length,  
    AVG(int_rate) AS avg_int_rate  
  FROM cte1  
  GROUP BY 1)  
  
  SELECT  
    CASE  
      WHEN issue_year IN (2015, 2017) THEN 'Others'  
      WHEN issue_year IN (2014, 2016) THEN 'Low TKB30'  
      WHEN issue_year IN (2018, 2019) THEN 'High TKB30'  
      END AS cohort_group,  
    ROUND(SUM(total_customer), 0) AS total_customer,  
    ROUND(AVG(avg_income), 0) AS avg_income,  
    ROUND(AVG(home_ownership / total_customer), 4) AS  
    home_ownership_rate,  
    ROUND(AVG(avg_enp_length), 2) AS avg_enp_length,  
    ROUND(AVG(avg_int_rate), 4) AS avg_int_rate  
  FROM cte2  
  GROUP BY 1
```

SQL – Q7 d

```
WITH cte1 AS (  
  SELECT  
    issue_year,  
    COUNT(c.customer_id) AS total_customer,  
    state,  
    COUNT(c.customer_id) OVER (ORDER BY issue_year) AS total_customer_per_state  
  FROM `nextproject-001.FIP.loan` l  
  LEFT JOIN `nextproject-001.FIP.customer` c  
  ON l.customer_id = c.customer_id  
  WHERE issue_year IN (2014, 2015, 2016, 2017, 2018, 2019)  
  GROUP BY 1, 3),
```

```
cte2 AS (SELECT  
  CASE  
    WHEN issue_year IN (2015, 2017) THEN 'Others'  
    WHEN issue_year IN (2014, 2016) THEN 'Low TKB30'  
    WHEN issue_year IN (2018, 2019) THEN 'High TKB30'  
  END AS cohort_group,  
  SUM(total_customer) AS sum_total_customer,  
  state,  
  total_customer / total_customer_per_state AS percentage_per_state  
  FROM cte1  
  GROUP BY 1, 3)
```

```
SELECT cohort_group, total_customer, state, percentage_per_state  
FROM cte2  
GROUP BY 1, 3
```

SQL – Q7 e

```
WITH cte1 AS (  
SELECT  
issue_year,  
COUNT(customer_id) AS total_customer,  
purpose  
FROM `nextproject-001.FIP.loan`  
WHERE issue_year IN (2014, 2015, 2016,  
2017, 2018, 2019)  
GROUP BY 1, 3)
```

```
SELECT  
CASE  
WHEN issue_year IN (2015, 2017) THEN  
'Others'  
WHEN issue_year IN (2014, 2016) THEN 'Low'  
TKB30  
WHEN issue_year IN (2018, 2019) THEN 'High'  
TKB30  
END AS cohort_group,  
SUM(total_customer),  
purpose  
FROM cte1  
GROUP BY 1, 3  
ORDER BY 1, 3
```